澳門理工大學
Universidade Politécnica de Macau
Macao Polytechnic University

FACULTY OF APPLIED SCIENCES

PHD OF ARTIFICIAL INTELLIGENCE EMPOWERED SMART HEALTHY AGEING

LEARNING MODULE OUTLINE

Academic Year	2026/2027	Semester	1
Module Code	AIHA8123		
Learning Module	Artificial Intelligence Driven Meta-Analysis		
Pre-requisite(s)	Nil		
Medium of Instruction	English		
Credits	3	Contact Hours	45 hrs
Instructor	Dr. Wenwang Rao	Email	raoww@mpu.edu.mo
Office	ACADEMIC BUILDING; E1303-2	Office Phone	8599 3826

MODULE DESCRIPTION

This course covers the core principles and methodologies of evidence-based healthcare (EBH) and meta-analysis, including systematic reviews, quantitative evidence synthesis, bias evaluation, and statistical methods for data combination. It also highlights AI-powered meta-analysis, which uses artificial intelligence to automate literature retrieval, data harmonisation, and quality appraisal, making evidence-based practice more efficient and precise.

MODULE INTENDED LEARNING OUTCOMES (ILOS)

On completion of this learning module, students will be able to:

M1.	Explain the basic concepts of meta-analysis; (APEP4-M1)
M2.	Conduct systematic reviews and meta-analyses using appropriate statistical software; (AHEP4-M2, AHEP4-M3, AHEP4-M13)
M3.	Critically appraise the quality of evidence and interpret the results of meta-analyses for clinical and policy decision-making; (AHEP4-M4, AHEP4-M12)
M4.	Synthesis and project in the form of a report document and presentation (AHEP4-M2, AHEP4-M16, AHEP4-M17).

These ILOs aims to enable students to attain the following Programme Intended Learning Outcomes (PILOs):

PILOs		**M1**	**M2**	**M3**	**M4**
P1.	Scientific methodologies and techniques of AI in meta-analysis and evidence synthesis;	√			
P2.	Knowledge and in-depth understanding of a wide range of meta-analysis-related topics;	√			

P3.	Knowledge and hands-on experience of analysis, assessment and solutions of data synthesis and meta-analysis-related issues;	√	√	√	√
P4.	Knowledge and application of AI-related methodologies in innovative solutions for literature screening and data extraction;		√		
P5.	Initiate original researches in AI-assisted systematic reviews and meta-analysis, both individually and collaboratively in a team;		√	√	√
P6.	Plan, design, execute and manage a scholarly research project;		√		√
P7.	Critically assess and analyse an advanced technical issue in large-scale data synthesis			√	
P8.	Communicate research findings, both orally to diverse audiences and in writing through publishing research papers of scholarly values;				√
P9.	Gather and disseminate knowledge at the postgraduate level and beyond;				√
P10.	To demonstrate advanced knowledge, competence and research capability in AI-driven meta-analysis;				√
P11.	To illustrate a global vision on knowledge advancement and dissemination;				√
P12.	To demonstrate professional integrity and the spirit of challenge;				√
P13.	To advocate professionalism in workplaces and the society at-large;				√
P14.	To communicate professionally and effectively both in speaking and in writing.				√

MODULE SCHEDULE, COVERAGE AND STUDY LOAD

Week	Content Coverage	Contact Hours
1-2	1. Introduction to Evidence-based Healthcare	6
	1.1 Definition and principles of EBH	
	1.2 Formulating clinical questions (PICO framework)	
	1.3 Types of clinical studies and their relevance to EBH	
3-6	2. Introduction to Meta-analysis and Systematic Reviews	12
	2.1 Overview of meta-analysis and systematic reviews	
	2.2 Protocol development and registration	
	2.3 Comprehensive literature search strategies	
	2.4 Data extraction and management	
	2.5 Quality assessment tools	
7	3. Statistical Methods for Meta-analysis	3
	3.1 Effect size measures	
	3.2 Heterogeneity assessment	
8	4. Bias and Publication Bias	3
	4.1 Sources of bias in studies	

	4.2 Assessing and addressing publication bias	
9	5. Interpreting and Reporting Meta-analysis Results	3
	5.1 Interpretation of meta-analysis findings	
	5.2 Guidelines for reporting (PRISMA)	
10	6. Software and AI Tools for Meta-analysis	3
	6.1 Introduction to meta-analysis software	
	6.2 Using R for meta-analysis	
11	7. Case Studies and Practical Applications	3
	7.1 Critical appraisal of published meta-analyses	
	7.2 Real-world examples and case studies	
12-14	8. Performing a Systematic Review and Meta-analysis	9
	8.1 conducting systematic reviews	
	8.2 performing meta-analytic pooling	
15	9. Student Presentations	3

TEACHING AND LEARNING ACTIVITIES

In this learning module, students will work towards attaining the ILOs through the following teaching and learning activities:

Teaching and Learning Activities	**M1**	**M2**	**M3**	**M4**
T1. Lectures	√	√	√	√
T2. In-class exercises	√	√	√	√

ATTENDANCE

Attendance requirements are governed by "Academic Regulations Governing PhD Degree Programmes of Macao Polytechnic University". Students who do not meet the attendance requirements for the learning module shall be awarded an 'F' grade.

ASSESSMENT

In this learning module, students are required to complete the following assessment activities:

Assessment Activities	**Weighting (%)**	**AHEP4 LOs**	**ILOs to be Assessed**
A1. Assignment / Classwork	25	APEP4-M1, APEP4-M2, APEP4-M12, APEP4-M13	M1, M3,
A2. Tests	25	APEP4-M1, APEP4-M2, APEP4-M3, APEP4-M4, APEP4-M12, APEP4-M13	M1, M2, M3,
A3. Project	50	APEP4-M1, APEP4-M2, APEP4-M3, APEP4-M4, APEP4-M12, APEP4-M13, APEP4-M16, APEP4-M17	M1, M2, M3, M4

The assessment will be conducted following the University's Assessment Strategy (see www.mpu.edu.mo/teaching_learning/en/assessment_strategy.php). Passing this learning module indicates that students will have attained the ILOs of this learning module and thus acquired its credits.

REQUIRED READINGS

There are no official texts for this module. Module notes are distributed in the class.

REFERENCES

1.Michael B, Larry H, et al., 2009. Introduction to Meta-Analysis, John Wiley & Sons, Ltd.

2.Higgins JPT, Thomas J, et al., 2019. Cochrane Handbook for Systematic Reviews of Interventions 2nd Edition, John Wiley & Sons, Ltd.

STUDENT FEEDBACK

At the end of every semester, students are invited to provide feedback on the learning module and the teaching arrangement through questionnaires. Your feedback is valuable for instructors to enhance the module and its delivery for future students. The instructor and programme coordinators will consider all feedback and respond with actions formally in the annual programme review.

ACADEMIC INTEGRITY

The Macao Polytechnic University requires students to have full commitment to academic integrity when engaging in research and academic activities. Violations of academic integrity, which include but are not limited to plagiarism, collusion, fabrication or falsification, repeated use of assignments and cheating in examinations, are considered as serious academic offenses and may lead to disciplinary actions. Students should read the relevant regulations and guidelines in the Student Handbook which is distributed upon the admission into the University, a copy of which can also be found at www.mpu.edu.mo/student_handbook/.